Siva Prasad Mandalapu

Irving, Texas, United States | msprasadm.1996@gmail.com | (845) 309-3962 | Github | Linkedin | Portfolio

PROFESSIONAL SUMMARY

AI/ML Engineer with 5 years of production experience in regulated healthcare and financial services. At Philips Healthcare, cut clinical inference latency from 2s to 150ms and deployed HIPAA-compliant federated learning across 5 hospital systems. At HDFC Bank, drove fraud detection to 95% recall while cutting losses 35% using transformer and graph neural network ensembles. AWS Certified ML Specialty; hands-on across PyTorch, MLOps, and cloud-scale deployment.

WORK EXPERIENCE

Philips Healthcare | AI/ML Engineer (Contract) Dec 2024 - Present

- Cut inference latency 13x (2s to 150ms per scan) at 10K+ scans/day with no loss in precision by converting PyTorch models to TensorRT and rebuilding model serving as containerized FastAPI services on on-prem GPU edge devices.
- Cut model release-candidate turnaround from 4 weeks to 1 by building ML CI/CD with GitHub Actions, Jenkins, Docker and Kubernetes, an MLflow model registry, and regression tests on a locked validation set.
- Monitored drift and model health for the 10K+ scans/day service with Prometheus and Grafana, tracking data drift, service health, and per-subgroup model performance to provide early visibility into model and service issues
- Made every training run reproducible over 500GB+ of imaging data with Spark and Kafka pipelines, schema validation and automated data-quality gates feeding AWS SageMaker training jobs.
- Trained models across 5 hospitals with zero patient images moved by building the client-side training and evaluation pipeline for a federated learning pilot (Flower).
- Helped cut radiologist review time 35% in an LLM pilot by building the LLMOps layer for a LangGraph report-drafting pipeline on Azure OpenAI: LangSmith tracing and automated checks for unsupported statements.

HDFC Bank | AI/ML Engineer, Fraud & Risk Dec 2019 - Nov 2022

- Served 5M+ real-time fraud scores/day at under 100 ms and 99.9% uptime by deploying containerized model endpoints on Azure ML and Kubernetes.
- Turned model updates into scheduled, automated releases by building Kubeflow retraining pipelines with MLflow experiment tracking and model versioning, feature and score drift detection, and automated rollback.
- Cut false-positive alerts 30% at 95% recall and fraud losses 35% by partnering with data scientists and fraud analysts to productionize an LSTM, Transformer and graph neural network ensemble.
- Cut fraud investigation time 40% by attaching SHAP explanations to every flagged transaction and auto-generating case summaries in the format required by RBI auditors
- Improved model F1 22% with Pandas and Dask feature engineering pipelines over 1M+ labeled transactions.

SKILLS

MLOps: CI/CD for ML, model registry, experiment tracking, automated retraining, drift monitoring, rollback, MLflow, Kubeflow, Airflow, GitHub Actions, Jenkins

Model Serving: real-time inference, GPU optimization, TensorRT, FastAPI, Docker, Kubernetes, AWS SageMaker, Azure ML

Observability: Prometheus, Grafana, LangSmith, Weights & Biases

Data & Cloud: Spark, Kafka, Pandas, Dask, PostgreSQL, AWS (S3, EC2, Lambda, EKS), Terraform

ML & Languages: Python, SQL, Bash, PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, LangGraph, SHAP, Flower

EDUCATION

Marist College Jan 2023 - Dec 2024
M.S., *Information Systems*

SIR CR REDDY College Of Engineering
B.Tech, *Mechanical Engineering*

CERTIFICATIONS & ACHIEVEMENTS

- AWS Certified Machine Learning – Specialty - AWS
- TensorFlow Developer Certificate (Google) - Google
- Generative AI with Large Language Models (DeepLearning.AI & AWS) - DeepLearning.AI & AWS